The PWDG–UWVF Equivalence Revisited

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Abstract

The ultra-weak variational formulation (UWVF) is the Trefftz-DG/PWDG scheme obtained for $\alpha = \beta = \delta = 1/2$. We explain these constants by an edgewise impedance identity: the two weighted PWDG jumps coincide with one half of the sum of the squared norms of the two oriented UWVF transmission defects. This identifies the TDG norm as their natural Riesz norm. In PWDG coordinates the skew-Hermitian part is iG, where G is the TDG Gramian, and Riesz normalization gives $K + iI$ with $K = K^*$. The discrete UWVF yields the complementary form $\frac{1}{2}I + iK_{\mathrm{U}}$. Thus the classical equivalence is a change between broken-field and impedance-trace coordinates for the same edge coupling.

Keywords: Trefftz methods, plane-wave discontinuous Galerkin, ultra-weak variational formulation, impedance traces, Riesz map, conditioning

1. Introduction

The ultra-weak variational formulation (UWVF) of Cessenat and Després uses impedance traces on element boundaries as unknowns [1, 2]. Plane-wave discontinuous Galerkin (PWDG), or more generally Trefftz-DG, instead uses broken Helmholtz fields and couples neighboring elements by numerical fluxes. The two methods have long been known to coincide for the particular choice

$$\alpha = \beta = \delta = \frac{1}{2}; \tag{1}$$

see [3, Remark 2.1] and [6, Sec. 2.2]. The usual derivation verifies this by inserting the UWVF fluxes into the DG formulation. It does not, however, make transparent why the factors 1/2 arise.

The point of this note is that (1) is forced by a simple impedance identity. The Dirichlet and normal-flux jumps used by PWDG are the two real channels of the complex transmission residual used by the UWVF. At the values (1), their weighted squares are exactly the norm of the two oriented impedance defects on each mesh edge. This also gives a direct Riesz interpretation of the Trefftz-DG norm. Since

$$\operatorname{Im} A_h(v, v) = \|v\|_{\mathrm{TDG}}^2 \tag{2}$$

for Trefftz fields [6], the skew-Hermitian part of the PWDG matrix is exactly i times the corresponding Gram matrix. Its inverse square root therefore produces a normal form canonical relative to the TDG Riesz metric, rather than merely rescaling an ill-conditioned plane-wave basis. The algebraic coincidence at (1) is classical, as is the TDG stability identity; the new observations here are the edgewise impedance isometry that explains why the three constants are one half and the paired PWDG–UWVF Riesz normal forms that result from it.

For direct comparison with Cessenat and Després we use their sign convention $\partial_n u + iku = g$. With the opposite time convention all impedance signs are obtained by replacing i with $-i$.

2. The two formulations

Let $\Omega \subset \mathbb{R}^2$ be polygonal and let $\mathcal{T}_h$ be a mesh with interior edges $\mathcal{F}_h^I$ and boundary $\Gamma = \partial\Omega$. We consider

$$-\Delta u - k^2 u = 0 \quad \text{in } \Omega, \qquad \partial_n u + iku = g \quad \text{on } \Gamma. \tag{3}$$

For an interior edge $e = \partial K^+ \cap \partial K^-$, with outward normals $n^\pm$, set

$$[\![v]\!]_N = v^+ n^+ + v^- n^-, \qquad [\![\nabla v]\!]_N = \nabla v^+ \cdot n^+ + \nabla v^- \cdot n^-,$$

and use the usual arithmetic averages $\{\{v\}\}$ and $\{\{\nabla v\}\}$. For each element let $V(K) \subset H^2(K)$ be a finite-dimensional Trefftz space,

$$V(K) \subset \{v : -\Delta v - k^2 v = 0 \text{ in } K\},$$

and set

$$V_h := \{v \in L^2(\Omega) : v|_K \in V(K) \text{ for every } K \in \mathcal{T}_h\}.$$

For PWDG, $V(K)$ is typically spanned by plane waves.

2.1. PWDG

The standard primal Trefftz-DG formulation [3, 6] is: find $u_h \in V_h$ such that

$$A_h(u_h, v_h) = \ell_h(v_h) \qquad \forall v_h \in V_h, \tag{4}$$

where, for $\alpha, \beta > 0$ and $0 < \delta < 1$,

$$\begin{aligned} A_h(u,v) = \int_{\mathcal{F}_h^I} \Big(&\{\{u\}\}\, [\![\nabla \bar v]\!]_N - \{\{\nabla u\}\} \cdot [\![\bar v]\!]_N + \alpha ik\, [\![u]\!]_N \cdot [\![\bar v]\!]_N \\ &- \beta (ik)^{-1} [\![\nabla u]\!]_N [\![\nabla \bar v]\!]_N \Big)\, ds \\ + \int_\Gamma \Big(&(1-\delta) iku\bar v + (1-\delta) u\, \overline{\partial_n v} - \delta\, \partial_n u\, \bar v \\ &- \delta (ik)^{-1} \partial_n u\, \overline{\partial_n v} \Big)\, ds, \end{aligned} \tag{5}$$

and

$$\ell_h(v) = \int_\Gamma g \Big((1-\delta) \bar v - \delta (ik)^{-1} \overline{\partial_n v} \Big)\, ds. \tag{6}$$

The associated Trefftz-DG norm is

$$\begin{aligned}\|v\|_{\mathrm{TDG}}^2 = k^{-1}\|\beta^{1/2}[\![\nabla v]\!]_N\|_{\mathcal{F}_h^I}^2 + k\|\alpha^{1/2}[\![v]\!]_N\|_{\mathcal{F}_h^I}^2 \\ + k^{-1}\|\delta^{1/2}\partial_n v\|_\Gamma^2 + k\|(1-\delta)^{1/2}v\|_\Gamma^2.\end{aligned} \tag{7}$$

For Trefftz functions, (2) follows directly from (5). For the impedance problem (3), (7) is a norm on the discrete Trefftz space: zero jumps and zero boundary contribution give a global homogeneous solution, hence zero by uniqueness.

2.2. UWVF

Let

$$\mathcal{X} = \prod_{K\in\mathcal{T}_h} L^2(\partial K).$$

For $y_K \in L^2(\partial K)$, let e_K solve the local adjoint-impedance problem

$$-\Delta e_K - k^2 e_K = 0 \text{ in } K, \qquad (-\partial_{n_K} + ik)e_K = y_K \text{ on } \partial K,$$

and define the local trace map

$$F_K y_K = (\partial_{n_K} + ik)e_K. \tag{8}$$

For the unit-impedance boundary condition in (3), the UWVF reads [1, 6]: find $x \in \mathcal{X}_h \subset \mathcal{X}$ such that, for all $y \in \mathcal{X}_h$,

$$\begin{aligned}\sum_{K\in\mathcal{T}_h}\int_{\partial K} x_K \bar{y}_K\, ds - \sum_{K\in\mathcal{T}_h}\sum_{K'\sim K}\int_{\partial K\cap\partial K'} x_{K'}\,\overline{F_K y_K}\, ds \\ = \sum_{K\in\mathcal{T}_h}\int_{\partial K\cap\Gamma} g\,\overline{F_K y_K}\, ds.\end{aligned} \tag{9}$$

Here $K' \sim K$ denotes the element across an interior edge and

$$\mathcal{X}_h = \{x \in \mathcal{X} : x_K = (-\partial_{n_K} + ik)v_h|_K,\ v_h \in V_h\}.$$

Our $\mathcal{X}_h$ is the discrete trace space denoted V_h in [1]; their wavenumber ω is our k. Their boundary parameter t is unrelated to the PWDG flux parameter δ; (3) is the case $t = 0$ of their boundary family. Following Cessenat and Després [1, Def. 1.5], $x_K = (-\partial_{n_K} + ik)v_h|_K$ is the outgoing trace and $F_K x_K = (\partial_{n_K} + ik)v_h|_K$ is the incoming trace. The local field is recovered from its outgoing trace. Gittelson, Hiptmair and Perugia showed that (4)–(6) with $\alpha = \beta = \delta = 1/2$ is precisely (9) written in broken-field variables [3, Remark 2.1].

3. Why the parameters are one half

Consider one interior edge $e = \partial K \cap \partial K'$, and abbreviate

$$u_K = u|_K, \quad u_{K'} = u|_{K'}, \quad q_K = \partial_{n_K} u_K, \quad q_{K'} = \partial_{n_{K'}} u_{K'}.$$

Introduce the normalized outgoing and incoming traces

$$\gamma_K^- u = \frac{-q_K + iku_K}{\sqrt{2k}}, \qquad \gamma_K^+ u = \frac{q_K + iku_K}{\sqrt{2k}}. \tag{10}$$

Thus γ_K^- is outgoing and γ_K^+ incoming in the terminology of [1, Def. 1.5]. The factor $\sqrt{2k}$ is conventional and cancels in the Riesz-normalized UWVF matrix. Exact transmission is $\gamma_K^- u = \gamma_{K'}^+ u$ and $\gamma_{K'}^- u = \gamma_K^+ u$. Define the two oriented defects

$$r_K = \gamma_K^- u - \gamma_{K'}^+ u, \qquad r_{K'} = \gamma_{K'}^- u - \gamma_K^+ u.$$

Proposition 1 (Impedance identity)**.** *For every broken field with well-defined traces on e,*

$$\frac{1}{2}\big(\|r_K\|_e^2 + \|r_{K'}\|_e^2\big) = \frac{1}{2k}\|q_K + q_{K'}\|_e^2 + \frac{k}{2}\|u_K - u_{K'}\|_e^2. \tag{11}$$

Thus the interior part of (7) for $\alpha = \beta = 1/2$ equals one half of the sum of the squared norms of the two oriented UWVF transmission defects. Equivalently, the map from the scaled value/flux jumps to $(r_K, r_{K'})$ is an isometry up to the fixed factor $\sqrt{2}$.

Proof. From (10),

$$r_K = \frac{-(q_K + q_{K'}) + ik(u_K - u_{K'})}{\sqrt{2k}}, \quad r_{K'} = \frac{-(q_K + q_{K'}) - ik(u_K - u_{K'})}{\sqrt{2k}}.$$

Use $|a+ib|^2 + |a-ib|^2 = 2|a|^2 + 2|b|^2$, valid for complex a, b, and integrate over e. □

The boundary factor is equally explicit. For either element-boundary trace in (10),

$$\frac{1}{2}\big(\|\gamma^- v\|_\Gamma^2 + \|\gamma^+ v\|_\Gamma^2\big) = \frac{1}{2k}\|\partial_n v\|_\Gamma^2 + \frac{k}{2}\|v\|_\Gamma^2, \tag{12}$$

which is exactly the boundary contribution in (7) for $\delta = 1/2$. More generally, replacing (10) by $-\sqrt{\beta/k}\, q_K + i\sqrt{\alpha k}\, u_K$ and its outgoing counterpart produces the jump weights β/k and αk; thus α/β selects the effective impedance $k\sqrt{\alpha/\beta}$. The same weighted identity extends to other positive PWDG flux parameters; on the exterior boundary the trace scaling is chosen consistently with the two weights δ/k and $(1-\delta)k$.

4. Riesz normal forms

Let $\{\Phi_j\}_{j=1}^N$ be a basis of V_h, and define the PWDG matrix and DG Gram matrix by

$$A_{j\ell} = A_h(\Phi_\ell, \Phi_j), \qquad G_{j\ell} = (\Phi_\ell, \Phi_j)_{\mathrm{TDG}}.$$

Thus, for $v = \sum_j c_j \Phi_j$, $c^* A c = A_h(v, v)$ and $c^* G c = \|v\|_{\mathrm{TDG}}^2$.

Theorem 1 (PWDG Riesz normal form)**.** *Assume that G is positive definite on the discrete Trefftz space. Then*

$$G = \frac{A - A^*}{2i}. \tag{13}$$

With $B = G^{-1/2}$, the Hermitian positive-definite inverse square root of G,

$$B^* A B = K + iI, \qquad K = K^*. \tag{14}$$

Hence the Riesz-normalized PWDG matrix is normal and its spectrum lies on $\operatorname{Im} z = 1$.

Proof. On an interior edge, with $q^\pm = \partial_{n^\pm} v^\pm$,

$$\operatorname{Im}(\{\{v\}\}\,[\![\nabla\bar v]\!]_N - \{\{\nabla v\}\}\cdot[\![\bar v]\!]_N) = \operatorname{Im}(v^+\overline{q^+} + v^-\overline{q^-}).$$

Together with the non-penalty boundary terms, the consistency contribution is $\operatorname{Im}\sum_K \int_{\partial K} v\overline{\partial_{n_K} v}\,ds = 0$, since Green's identity makes each integral $\int_K(|\nabla v|^2 - k^2|v|^2)\,dx$ real. The four remaining penalty terms give exactly (7), hence (13) by polarization. Writing $A = H + iG$, $H = H^*$, and using the Hermitian inverse square root $B = G^{-1/2}$, for which $B^*GB = I$, gives (14). □

The normal form is basis independent in the following precise sense. If $Hx = \mu Gx$, then the μ's are unchanged by any nonsingular change of Trefftz basis, while the normalized eigenvalues are $\mu + i$ and the singular values are $\sqrt{1+\mu^2}$. Thus the raw $\kappa_2(A)$, which can vary drastically under rescaling or a nearly redundant plane-wave dictionary, is separated from the intrinsic conditioning of the variational operator in its DG Riesz metric. If $G = V\Lambda V^*$ has tiny eigenvalues and $B_\tau = V_r\Lambda_r^{-1/2}$ is formed from the retained modes, then $B_\tau^* A B_\tau = K_\tau + iI_r$ on the compressed subspace.

For UWVF, [1, eqs. (2.21)–(2.23)] gives $(D - C)X = b$, where D is the trace Gram matrix. After its orthonormalization,

$$A_{\mathrm{U}} = D^{-1/2}(D-C)D^{-1/2} = I - U_h, \tag{15}$$

where $U_h = D^{-1/2}CD^{-1/2}$ is similar to their iteration matrix $D^{-1}C$. It is contractive in the present absorbing case (cf. [1, Lemma 1.12, Prop. 1.10]), hence

$$G_{\mathrm{U}} := A_{\mathrm{U}} + A_{\mathrm{U}}^* = (I - U_h)^*(I - U_h) + (I - U_h^*U_h) \ge 0. \tag{16}$$

Whenever $G_{\mathrm{U}} > 0$, $B_{\mathrm{U}} = G_{\mathrm{U}}^{-1/2}$ gives

$$B_{\mathrm{U}}^* A_{\mathrm{U}} B_{\mathrm{U}} = \frac{1}{2}I + iK_{\mathrm{U}}, \qquad K_{\mathrm{U}} = K_{\mathrm{U}}^*. \tag{17}$$

The first term in (16) measures transmission mismatch and the second measures dissipation caused by the absorbing boundary or by discrete compression. In the lossless invariant case U_h is unitary, the second term vanishes, and $U_h q = e^{i\theta}q$ implies $G_{\mathrm{U}} q = 4\sin^2(\theta/2)q$, the Hermitian-part counterpart of the unit-disk spectral picture in [1, Lemma 2.5].

5. A small numerical check

We use the unit square divided into two triangles, $k = 4$, and p equispaced plane waves on each element, centered at its centroid. All edge integrals use 80-point Gauss–Legendre quadrature. To show that the Riesz normalization is not tied to the UWVF parameters, the PWDG matrix is assembled from (5) with

$$\alpha = 1, \qquad \beta = 2, \qquad \delta = \frac{1}{4},$$

whereas the UWVF uses its standard impedance traces. The UWVF diagonal blocks are outgoing-trace Gram matrices, while an interior-edge block pairs the neighboring outgoing trace with the incoming test trace. Set

$$\widehat{A}_{\mathrm{P}} = G_{\mathrm{P}}^{-1/2} A_{\mathrm{P}} G_{\mathrm{P}}^{-1/2}, \qquad \widehat{A}_{\mathrm{U}} = G_{\mathrm{U}}^{-1/2} A_{\mathrm{U}} G_{\mathrm{U}}^{-1/2}.$$

Table 1: Raw and Riesz-normalized condition numbers.

p	$\kappa_2(A_\mathrm{P})$	$\kappa_2(\widehat{A}_\mathrm{P})$	$\kappa_2(A_\mathrm{U})$	$\kappa_2(\widehat{A}_\mathrm{U})$
5	1.23×10^1	1.225	4.18	1.192
9	3.33×10^2	1.264	1.81×10^2	1.374
13	2.48×10^5	1.392	9.12×10^4	1.755
17	7.73×10^8	1.634	3.05×10^8	2.379

For PWDG the assembled matrices satisfy

$$\frac{\|(A_\mathrm{P} - A_\mathrm{P}^*)/(2i) - G_\mathrm{P}\|_F}{\|G_\mathrm{P}\|_F} \le 1.5 \times 10^{-16}$$

over the tested rows. Raw conditioning grows by many orders of magnitude while the Riesz-normalized values stay below 2.4. The defining identities are edgewise, so the two-element mesh is only a compact illustration.

6. Discussion

The classical equality of PWDG and UWVF at $\alpha = \beta = \delta = 1/2$ is therefore an isometry statement: the value and normal-flux channels become the oriented impedance defects. Their Riesz normal forms are complementary, $K + iI$ in PWDG and $\frac{1}{2}I + iK_\mathrm{U}$ in UWVF. This connects with operator preconditioning [4] and UWVF boundary orthonormalization [5], but here the PWDG Riesz matrix is assembled directly from the jump and boundary terms already present in the TDG stability norm, not from an auxiliary mass matrix. Under a basis change S, $A \mapsto S^*AS$ and $G \mapsto S^*GS$, whereas the pencil (H, G) and Riesz-normalized spectrum are invariant; this separates representation conditioning from the intrinsic discrete operator.

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